Team Details

- ➔ Team name – Keyboard Smasher
- ➔ Team leader name – Dev Yagnik
- ➔ Problem Statement - Community partnerships (NSS/NGOs) for pilots and feedback.

Brief about the idea

➔ HumanLoop is an AI-powered social intelligence platform designed to intelligently connect individuals, donors, and volunteers with the most relevant NGOs.

➔ The platform leverages a Retrieval-Augmented Generation (RAG) architecture to transform fragmented and unstructured NGO data into a structured, searchable, and context-aware knowledge system.

➔ Instead of functioning as a static directory, HumanLoop enables users to submit real-world queries in natural language. The system retrieves verified NGO information, processes it through a contextual AI pipeline, and generates accurate, grounded, and actionable recommendations.

➔ HumanLoop bridges the critical gap between social need and effective action, ensuring that intent is efficiently translated into measurable impact.

Opportunities

➔ **How different is it from any of the other existing ideas?**

HumanLoop is not just an NGO directory or a simple project tracker. It is an end-to-end AI-powered Social Pilot Management Platform that combines intelligent planning, smart NGO matching, integrated payments, and real-time execution tracking in one ecosystem.

Unlike existing solutions that focus on only discovery or only funding, HumanLoop manages the entire lifecycle of a social initiative — from idea generation to certification — powered by a RAG-based AI engine and location-aware scoring system.

How will it be able to solve the problem?

HumanLoop addresses fragmentation in the social impact ecosystem by:

- Using AI to automatically generate structured pilot plans
- Matching NGOs based on location proximity and domain relevance
- Enabling secure funding through Razorpay and Stripe
- Providing expense transparency and milestone-based tracking
- Offering multilingual access for beneficiaries

This ensures transparency, accountability, and efficient execution — reducing failure rates of social pilots.

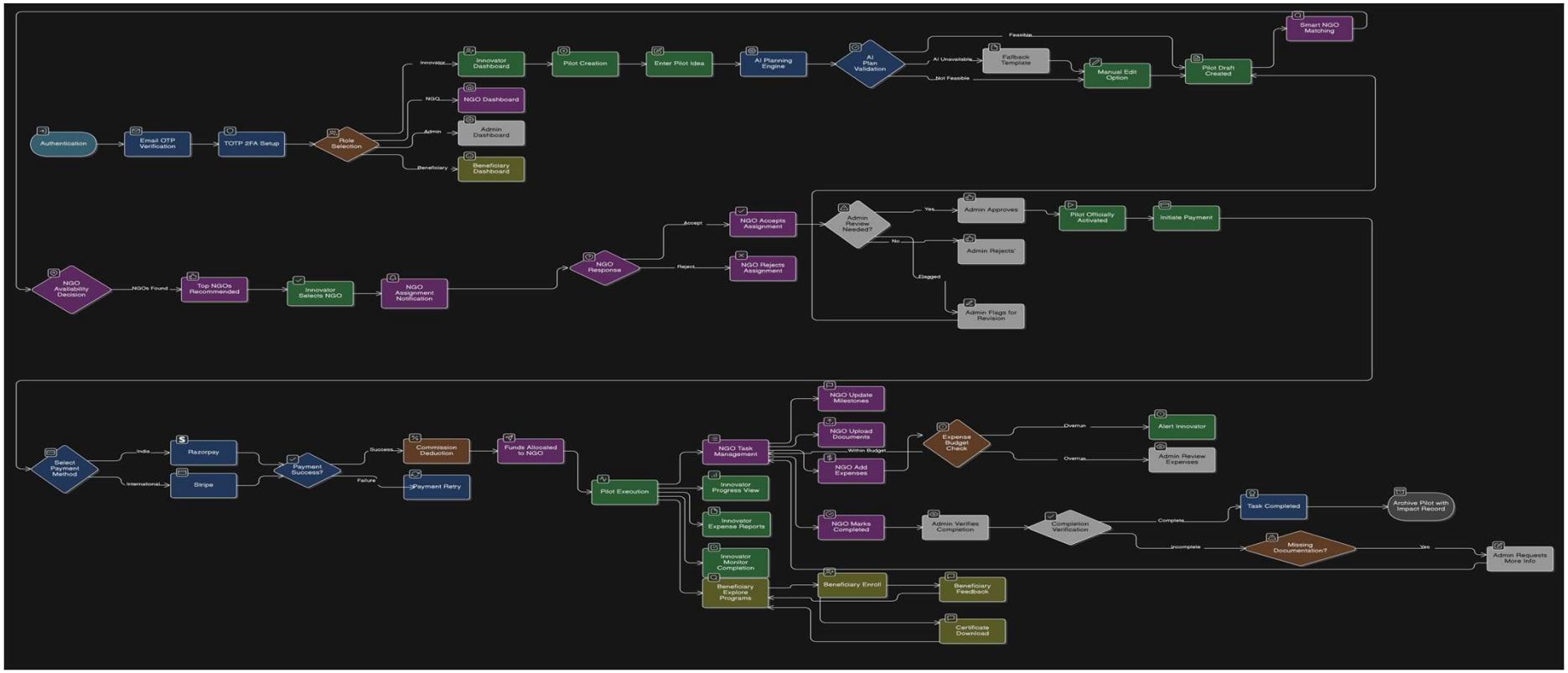
USP of the proposed solution

- End-to-end Social Pilot Lifecycle Management
- AI-powered Planning + Smart NGO Matching (RAG + FAISS)
- Dual Payment Gateway with built-in platform commission
- Role-based dashboards for all stakeholders
- Real-time tracking, document vault, and certification system
- Scalable architecture adaptable for CSR, government, and global expansion

List of features offered by the solution

- **Role-Based Dashboards**
  Dedicated interfaces for Innovator, NGO, Admin, and Beneficiary.
- **AI-Powered Pilot Planning**
  Automatically generates structured pilot plans with estimated team size and beneficiary outreach.
- **Smart NGO Matching**
  Location-aware and domain-based scoring system for optimal NGO selection.
- **RAG-Based AI Engine**
  Context-aware recommendations using Ollama + FAISS for intelligent and reliable decision support.
- **Integrated Dual Payment Gateway**
  Secure transactions via Razorpay and Stripe with automated platform commission handling.
- **Expense Tracking & Transparency**
  NGOs manage expenses; Innovators have real-time read-only financial visibility.
- **Task & Milestone Management**
  Checklist-driven execution tracking to ensure accountability and completion.
- **Real-Time Notifications**
  Instant alerts for approvals, assignments, payments, and milestones.
- **12-Language Multilingual Support**
  Inclusive beneficiary access across 12 regional languages to ensure wider reach and digital accessibility.
- **Document Vault & Certificate Generation**
  Secure file management with downloadable completion certificates (PDF export).
- **Two-Factor Authentication (2FA)**
  TOTP-based verification system for enhanced platform security.

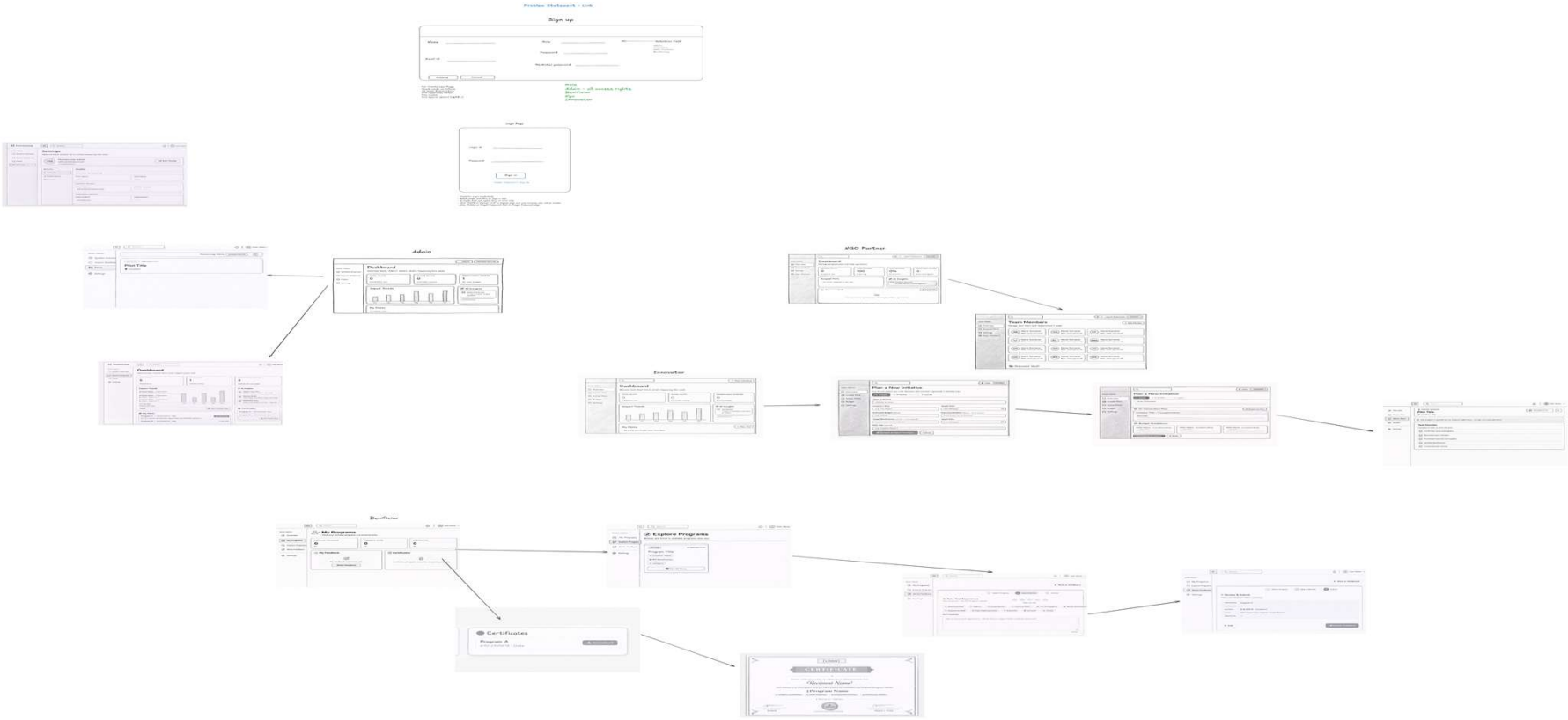
Process flow diagram or Use-case diagram



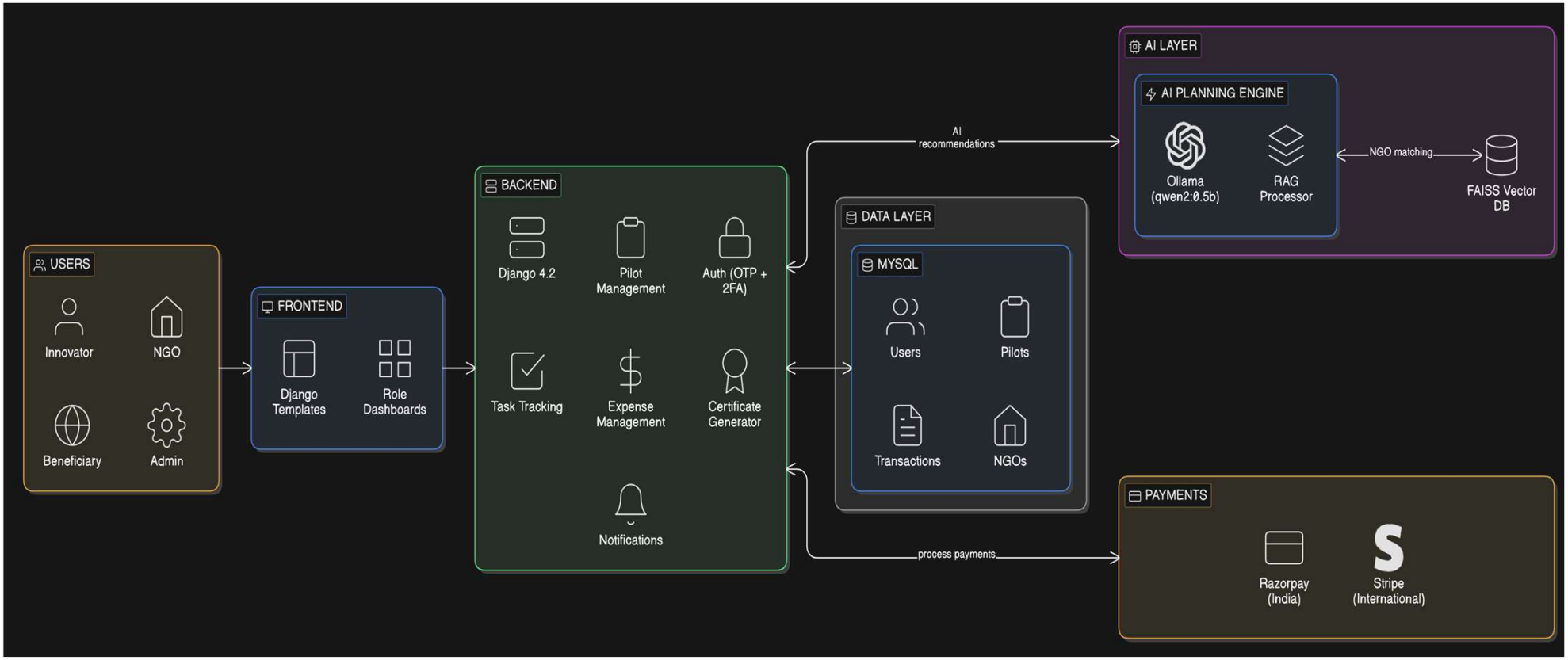
Wireframes/Mock diagrams of the proposed solution (optional)

To see content clearly pls click on link ➔

https://excalidraw.com/#room=5410ec6f8c05c909306b,TPW0c988JkHUF8tJTTm41g



Architecture diagram of the proposed solution



Technologies to be used in the solution

➔ Programming Language
Python 3.10+ – Core language powering backend logic and AI integration.

➔ Backend Framework
Django 4.2.28 – Manages authentication, role-based access, pilot lifecycle, dashboards, and APIs.

➔ Database Layer
MySQL 8.0+ – Structured relational database
mysqlclient & PyMySQL – Database connectivity drivers

➔ AI & Intelligence Layer
Ollama (qwen2:0.5b) – AI-powered pilot planning
FAISS – Vector-based smart NGO matching
Requests – API communication for AI services

➔ Payment Integration
Razorpay (2.0.0) – Indian payments
Stripe (14.4.0) – International payments

➔ Authentication & Security
pyotp – TOTP-based Two-Factor Authentication
qrcode – QR generation for 2FA setup
Email OTP verification system
python-dotenv – Secure environment variable management

➔ Frontend Technologies
HTML, CSS, JavaScript (Django Templates)

➔ Production & Deployment
Gunicorn (23.0.0) – WSGI production server
WhiteNoise (6.9.0) – Static file serving (Railway/Production)

➔ File & Media Handling
Pillow (12.1.1) – Image processing for uploads and certificates

Usage of AMD Products/Solutions

Our website was developed and tested on a system powered by Advanced Micro Devices (AMD) processor (Ryzen 7 6000 Series).

The AMD processor was used to run the development environment, local server, and database during project implementation.

It handled multitasking efficiently, including coding, testing, debugging, and browser execution simultaneously.

The high-performance multi-core architecture ensured smooth execution and faster processing of our AI-based application.

AMD technology contributed to reliable performance and efficient development of our web platform.

Estimated implementation cost (optional)

➔ **Cloud Infrastructure (6 Months)**
₹75,000 – ₹1,20,000
(Hosting, database, storage, backups)

➔ **AI Compute & Processing**
₹30,000 – ₹60,000
(Ollama inference, vector processing)

➔ **Payment Gateway Charges**
2%–3% per transaction (Razorpay / Stripe)
(No fixed setup cost)

➔ **Development Cost (Market Equivalent)**
₹4–6 Lakhs
(Currently built by team)

Prototype Assets (Optional)

- **GitHub Public Repository Link**
- https://github.com/DevNovaOps/Humanloop
- Demo Video Link (Max: 3 Minutes)
- The below video is 2.4x speed of original video to make it 3 Minute video:-
- https://drive.google.com/file/d/1Iye6bM85fN6X_gKvkLWukgYSULVogB2U/view?usp=drive_link
- This below video is original video of 7 minute:-
- https://drive.google.com/file/d/12M0uCxSDVMIbgzCP2eP9hKZS8pCCSQ_J/view?usp=drive_link

AMD Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by H2S HACK2SKILL

Thank you!